# Harshit Chaudhary

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## Experience

**Financial Controller** Jul 2026 - Present

*UBS* *Mumbai, India*

- Developing a unified Python/Flask application centralizing 15+ financial workflows, enabling business users to execute recurring processes through a streamlined interface.
- Integrating multiple external REST APIs to retrieve and transform market data using financial methodologies, generating structured outputs and analytics.
- Developing interactive dashboards and AI-powered automation solutions using Power BI and Python to identify trends and streamline recurring workflows.

**Automation Intern** Jan 2026 - Jun 2026

*UBS* *Mumbai, India*

- Constructed a Python-based automation pipeline aggregating 10+ Excel sources into 8-sheet structured reports, reducing manual effort by 30+ hours
- Formulated modular OOP workflows using Pandas, NumPy, and Openpyxl for data ingestion, validation, and transformation with robust error handling; introduced real-time logging for non-technical usability.
- Launched a Streamlit frontend to simplify interaction and prototyped an AI-driven chatbot for automated output analysis; assembled Power BI dashboards for visualization.

**Software Development Intern** | Link Jun 2025 - Jul 2025

*NIIT Managed Training Services* *Gurugram, India*

- Created a full-stack user management system with React, Node/Express, PostgreSQL, and JWT-based authentication.
- Deployed the application on Vercel with a cloud backend, refining database queries to achieve sub-800ms response times.
- Contributed 50% of backend logic and 40% of UI features within Agile sprints alongside a 5-member team.

## Projects

**Smart Stick for the Visually Impaired** | Link | *Arduino, ML, Computer Vision* Dec 2024 – Apr 2025

- Engineered an Arduino-based assistive device integrating ultrasonic sensors, GPS navigation, and camera-based object detection with haptic feedback.
- Devised ML-driven path optimization using Google Maps APIs to avoid crime, accident, and traffic hotspots, with SOS alerts triggered on anomalies or unfamiliar faces.

**AI-Based Frame Interpolation for WMS Services** | Link | *Python, ConvLSTM* Sep 2024 – Dec 2024

- Designed an end-to-end ML pipeline for frame interpolation to generate smooth satellite imagery videos for WMS platforms, enhancing temporal resolution of cloud movement visualizations.
- Integrated environmental parameters (wind, pressure, temperature) into a ConvLSTM model with a custom loss function to improve prediction of deformable patterns like cyclones.

**TrackKar – Dynamic Route Rationalization** | Link | *Python, ML, Flask, Folium, JS* Sep 2024 – Oct 2024

- Enhanced 50+ public routes using ML on GPS data, improving arrival-time accuracy by 18%.
- Processed 200+ GPS records/min and deployed real-time dashboards using Folium and Google Maps APIs, enabling adaptive scheduling via REST APIs.

**Saavdhaan – Crime Hotspot Detection** | Link | *Python, REST APIs, GeoPandas* Nov 2023 – Dec 2023

- Analyzed 15K+ crime records using clustering models (DBSCAN/KMeans) to identify hotspots with 80% accuracy.
- Deployed geofencing alerts (500m radius) and safer route recommendation algorithms for end-users.
- Streamlined ETL pipelines and indexing, reducing query times by 40%.

**Disaster Management Platform** | Link | *React, Flutter, Python* Sep 2023 – Nov 2023

- Built a cross-platform emergency response system tested by 200+ users in campus drills, integrating offline guides, 10+ helpline APIs, and multilingual chatbot alerts.
- Orchestrated resource mapping and notification workflows, reducing emergency response delays by 30%.

## Skills

**Languages:** JAVA, SQL, Python, C/C++, R, HTML, CSS, JavaScript, Dart
**Frameworks/Libraries:** React, Node.js, Express, Flask, Flutter, TensorFlow
**Databases/Cloud:** MySQL, PostgreSQL, MongoDB, AWS, GCP, Vercel, Google Colab
**Tools & Platforms:** Linux (Ubuntu), JWT, REST APIs, Docker, Kubernetes, Postman, Figma, Test Automation Basics, Git, CI/CD (GitHub Actions / GitLab CI)
**Core Competencies:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), System Design Basics, Distributed System Basics, Machine Learning Pipelines, Agentic & Generative AI

## Education

**Vellore Institute of Technology** Sep 2022 - Jul 2026
*B.Tech in Computer Science and Engineering (Artificial Intelligence and Robotics)* *CGPA: 8.81/10*

## Leadership Experience

**President, Dramatics Club** May 2024 - May 2025
*Vellore Institute of Technology, Chennai* *Chennai, India*

- Led 50+ members across 10+ large-scale performances, attracting 1K+ attendees and fostering collaboration.
- Streamlined logistics and budgeting workflows, reducing prep cycles by 50% and boosting workshop participation.
- Directed and scripted plays winning national accolades across IIT Madras, IIT Bombay, and VIT campuses.

## Achievements and Awards

- Secured **Top 5 National Team** at **Hackout 2023**, IIT Madras (1K+ participants).
- Selected for **Smart India Hackathon** 2023 & 2024 (top 1% campus-wide).
- Solved **400+ LeetCode problems**, achieving a 70% acceptance rate and strengthening DSA fundamentals.
- **First Prize**, Nukkad Natak at **Riviera 2025** (VIT Vellore); **Third Prize** in 2023 & 2024.
- **Finalist** at **Aagaaz 2024 (IIT Bombay)** nationwide street play contest.
- **Best Writer & Director** at **Saarang 2025 (IIT Madras)** and **Drama League 2024 (VIT Chennai)**.